

ANDRE KE

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🌐 <https://github.com/andrxke> 🌐 <https://andrxke.github.io>

Technical Skills

Languages & Technologies: Python, Java, C++, JavaScript, TypeScript, HTML/CSS, SQL, React, Express, Next.js, Tailwind CSS, MongoDB, Mongoose, RESTful APIs

Tools & Frameworks: Docker, Git, GitHub, Figma, Azure DevOps, Supabase, Pandas, ParaView, ITK-SNAP, Pytorch

Experience

Software Developer

Jan 2026 – April 2026

The Hospital for Sick Children (SickKids)

Toronto, ON

- Spearheaded the development, **Docker**-based containerization, and deployment of a **Django** clinical dashboard, streamlining the delivery of Plotly.js metrics and LLM research results to institutional researchers.
- Engineered a high-performance backend capable of processing **300 million gene data entries** utilizing **SQLite** and H5 files, integrating automated statistical tests on uploaded **RNA data** to correlate model predictions.
- Extended a **Google ADK**-based multi-agent system by engineering 3 modular tools and a custom library integration, enabling the agent to autonomously retrieve and process high-throughput genomic data with increased precision
- Engineered **bash** scripts to automate genetic workflows, delivering a 6x speedup in processing efficiency compared to manual workflows.

Teaching and Research Assistant

May 2025 – Aug 2025

University of Waterloo

Waterloo

- Developed an **image processing pipeline** in **Python** using **NumPy** and **Scikit-image** to detect potentially cancerous tissue in brain MRI scans.
- Created **3D "glass brain" visualizations** by converting **NIfTI (.nii)** files into **VTK surface meshes** with **ITK-SNAP** and rendering in **ParaView**.
- Authored 8+ new assignments and mentored over **250 students** for **Data Structures and Algorithms**, leading tutorials on **dynamic programming** and **graph algorithms**.

Software Developer

Sep 2024 – Dec 2024

Hire Harmoni

Remote

- Developed the front end of a SaaS application using **TypeScript**, **Next.js**, and **Tailwind CSS** while integrating a **Supabase** backend.
- Converted **Figma** designs into 15+ responsive pages and a library of **reusable components**, collaborating with the design team to refine the user experience.
- Worked with multiple **APIs** to handle authentication, data fetching, and real-time updates to support application functionality.

Software Quality Assurance Specialist

Sep 2024 – Dec 2024

Siemens Healthineers

Ottawa, ON

- Executed **system, integration, and validation testing** across **embedded systems**, Android, and web-based applications, ensuring **regulatory compliance** for medical devices.
- Developed **automated test scripts** in **Python** with **Appium**, resulting in a **5% increase** in test coverage within **Agile Sprints** on **Azure DevOps**.
- Explored the implementation of a virtual **Bluetooth stack** using the **Google Bumble** library and **Websockets** in an **Azure Cloud** environment.

Projects

ESG Analyzer | *React, TypeScript, Next.js, Tensorflow*

Winter 2024

- Developed a full-stack tool using **Next.js** and **TypeScript** to estimate **ESG scores** by gathering web content via Cheerio and experimenting with **TensorFlow.js (LSTM)** for real-time sentiment analysis.

Extracurricular

Vision and Image Processing Lab

Fall 2025

Undergraduate Research Assistant

University of Waterloo

- Enhanced **3D U-Net** performance for MRI segmentation through transfer learning and custom preprocessing; investigated the efficacy of VisionLLMs (SAM3) in clinical imaging tasks.

Education

University of Waterloo

Sep. 2023 – May 2028

*Candidate for Bachelor of Applied Science in **Systems Design Engineering***

Waterloo, ON